Character sheaves and modular representations of finite reductive groups

On the unitriangularity of their decomposition matrices

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Introduction and motivation

What are groups for?

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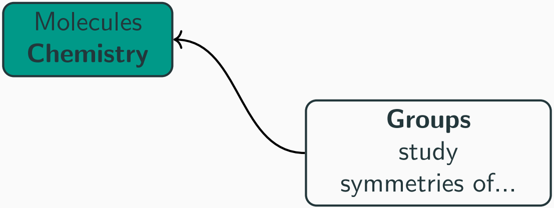
What are groups for?



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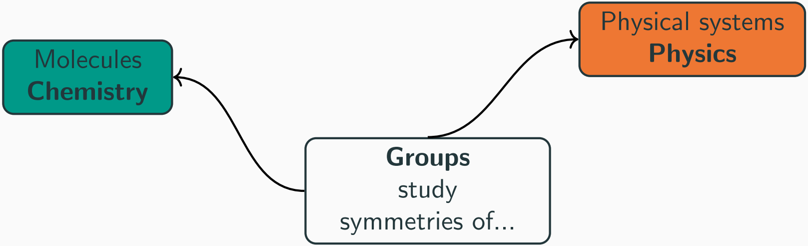

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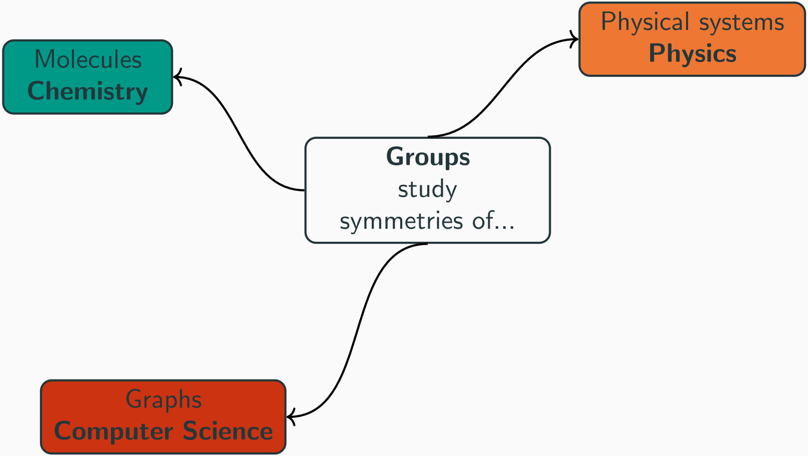

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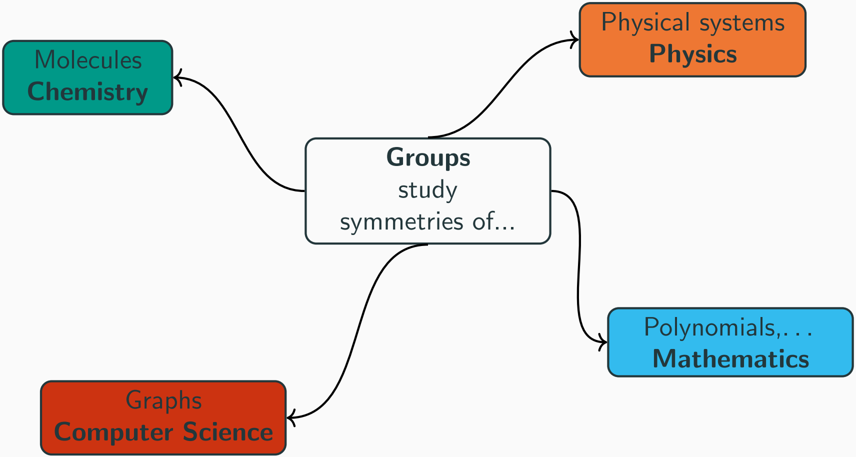

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Few constraints $\implies$ many difficult objects.

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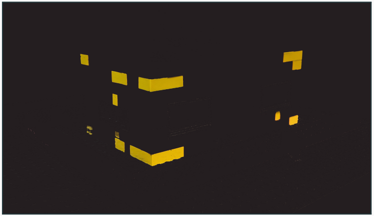

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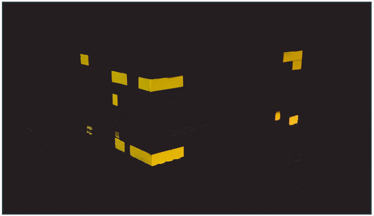

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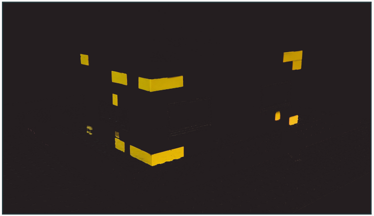

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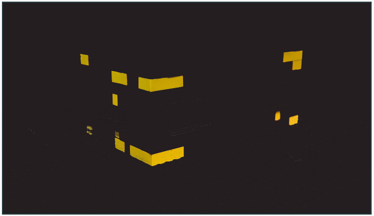

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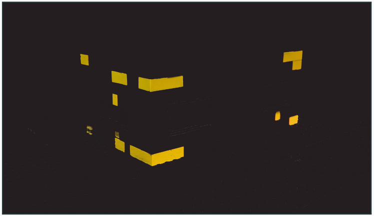

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Link? **Decomposition matrices**

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If $\mathrm{char}(\mathbb{F}) = \ell > 0$, we are in the **modular** case.

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Definition

The **ℓ-decomposition matrix** $D = (D_{V,W})_{V \in \mathrm{Irr}_{\mathbf{K}}(G), W \in \mathrm{Irr}_{\overline{\mathbb{F}_\ell}}(G)}$ is given by

$$D_{V,W} := \dim \left(\mathrm{Hom}_{\overline{\mathbb{F}_\ell}[G]}(V^{\mathcal{O}} \otimes_{\mathcal{O}} \overline{\mathbb{F}_\ell}, W) \right).$$

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$$1 \to V_1^{\mathcal{O}} \otimes_{\mathcal{O}} \overline{\mathbb{F}_3} \to V_3^{\mathcal{O}} \otimes_{\mathcal{O}} \overline{\mathbb{F}_3} \to V_2^{\mathcal{O}} \otimes_{\mathcal{O}} \overline{\mathbb{F}_3} \to 1.$$

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Thus, the 2-decomposition matrix of S_3 is $D = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$.

Now take $\ell = 3$.
The modules $V_1^{\mathcal{O}} \otimes_{\mathcal{O}} \overline{\mathbb{F}_3}$ and $V_2^{\mathcal{O}} \otimes_{\mathcal{O}} \overline{\mathbb{F}_3} \in \mathrm{Irr}_{\overline{\mathbb{F}_3}}(G)$ are irreducible and there is a short exact sequence:

$$1 \to V_1^{\mathcal{O}} \otimes_{\mathcal{O}} \overline{\mathbb{F}_3} \to V_3^{\mathcal{O}} \otimes_{\mathcal{O}} \overline{\mathbb{F}_3} \to V_2^{\mathcal{O}} \otimes_{\mathcal{O}} \overline{\mathbb{F}_3} \to 1.$$

Thus, the 3-decomposition matrix of S_3 is $D = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$.

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- $P^{\mathcal{O}} \otimes_{\mathcal{O}} \mathbf{K}$ a $\mathbf{K}[G]$-module,
- $P^{\mathcal{O}}$ an $\mathcal{O}[G]$-lattice s.t. $P^{\mathcal{O}} \otimes_{\mathcal{O}} \overline{\mathbb{F}_\ell} \cong P$,
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Definition (Decomposition matrices 2)

The ℓ-decomposition matrix $D = (D_{V,W})_{V \in \mathrm{Irr}_{\mathbf{K}}(G), W \in \mathrm{Irr}_{\overline{\mathbb{F}_\ell}}(G)}$ is given by:

$$P_W^{\mathcal{O}} \otimes_{\mathcal{O}} \mathbf{K} = \bigoplus_{V \in \mathrm{Irr}_{\mathbf{K}}(G)} D_{V,W} V = \bigoplus_{V \in \mathrm{Irr}_{\mathbf{K}}(G)} \langle V, P_W^{\mathcal{O}} \otimes_{\mathcal{O}} \mathbf{K} \rangle V.$$

Decomposition matrices: unitriangularity

Idea behind decomposition matrices

Ordinary modules + decomposition matrices $\approx$ modular modules.

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Example

If $G = S_3$, the 3-decomposition matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$ is unitriangular.

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Conjecture (Geck 1990)

The decomposition matrix of a finite reductive group in non-defining characteristic is lower-unitriangular (w.r.t. a suitable ordering).

Finite reductive groups and their representation theory

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Importance

The majority of finite simple groups are built of finite reductive groups.

Classification

Finite reductive groups are classified thanks to combinatorial data, the **complete root data**, which contain the root system.

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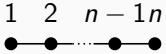
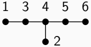
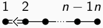
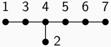
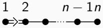
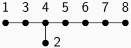
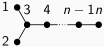
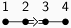
Classical types		Examples	Exceptional types		Examples
A_n $_{n\geq 1}$		$\mathrm{SL}_n(q)$, $\mathrm{GU}_n(q)$	E_6		$E_6(q)$, ${}^2E_6(q)$
B_n $_{n\geq 2}$		$\mathrm{Spin}_{2n+1}(q)$, $\mathrm{SO}_{2n+1}(q)$	E_7		$(E_7)_{\mathrm{sc}}(q)$, $(E_7)_{\mathrm{ad}}(q)$
C_n $_{n\geq 3}$		$\mathrm{Sp}_{2n}(q)$	E_8		$E_8(q)$
D_n $_{n\geq 4}$		$\mathrm{Spin}^+_{2n}(q)$, $\mathrm{SO}^+_{2n}(q)$	G_2		$G_2(q)$, ${}^2G_2(3^{2f+1})$
			F_4		$F_4(q)$, ${}^2F_4(2^{2f+1})$

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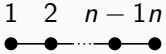
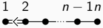
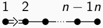
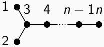
A_n $n \geq 1$		$\mathrm{SL}_n(q)$, $\mathrm{GU}_n(q)$
B_n $n \geq 2$		$\mathrm{Spin}_{2n+1}(q)$, $\mathrm{SO}_{2n+1}(q)$
C_n $n \geq 3$		$\mathrm{Sp}_{2n}(q)$
D_n $n \geq 4$		$\mathrm{Spin}^+_{2n}(q)$, $\mathrm{SO}^+_{2n}(q)$

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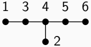
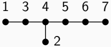
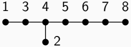
E_6		$E_6(q)$, ${}^2E_6(q)$
E_7		$(E_7)_{\mathrm{sc}}(q)$, $(E_7)_{\mathrm{ad}}(q)$
E_8		$E_8(q)$
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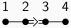
F_4		$F_4(q)$, ${}^2F_4(2^{2f+1})$

Ordinary representations

Fix an F-stable Borel subgroup $\mathbf{B} \leq \mathbf{G}$ and an F-stable maximal torus $\mathbf{T} \leq \mathbf{B}$. We write $(\mathbf{G}^*, F^*)$ for the pair in duality with $(\mathbf{G}, F)$.

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Theorem (Deligne–Lusztig 1976)

There is a partition

$$\mathrm{Irr}_{\mathbf{K}}(\mathbf{G}^F) = \bigsqcup_s \mathscr{E}(\mathbf{G}, s),$$

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The series $\mathscr{E}(\mathbf{G}, 1)$ is called the **unipotent** series.

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Fix a prime number $\ell \neq p$. Assume that F is a Frobenius map. Let $s \in (\mathbf{G}^*)^{F^*}$ be a semisimple ℓ'-element. Define

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The ℓ-decomposition matrix of the unipotent ℓ-blocks of $\mathbf{G}^F$ is lower-unitriangular if p is good for $\mathbf{G}$ and ℓ is very good for $\mathbf{G}$ (good + condition related to $Z(\mathbf{G})$).

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Strategy and results

Our setting

We fix:

- $\mathbf{G}$ a connected reductive **simple** group of **adjoint** type defined over $\overline{\mathbb{F}_p}$ for p a **good** prime for $\mathbf{G}$,
- $F : \mathbf{G} \to \mathbf{G}$ a **Frobenius** map,
- $\ell \neq p$ a prime number,
- $(\mathbf{G}^*, F^*)$ in duality with $(\mathbf{G}, F)$,
- $(\mathbf{K}, \mathcal{O}, \overline{\mathbb{F}_\ell})$ a splitting ℓ-modular system for $\mathbf{G}^F$ and
- $\mathscr{B} := \mathscr{E}_\ell(\mathbf{G}, s)$ for a semisimple F^*-stable ℓ'-element $s \in \mathbf{G}^*$.

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Goal

Check whether the ℓ-decomposition matrix of $\mathbf{G}^F$ restricted to $\mathscr{B}$ is lower-unitriangular.

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Theorem (Geck–Hiss 1991)
Assume that ℓ is good. Let $s \in \mathbf{G}^*$ be a semisimple F^*-stable ℓ'-element. Then the series $\mathscr{E}(\mathbf{G}, s)$ forms a basic set for $\mathscr{E}_\ell(\mathbf{G}, s)$.

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Step 3: generalised Gelfand–Graev representations

To each unipotent conjugacy class C of $\mathbf{G}$, one associates n_C projective modules Γ_i^C for $1 \leq i \leq n_C$, called the **generalised Gelfand–Graev representations** (GGGRs).

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This theorem is only known when p is **good**.

Step 3: Kawanaka characters

$\mathcal{C}_1$ $\mathcal{C}_2$ $\mathcal{C}_k$

Fix an order on the k unipotent conjugacy classes $C_1, \ldots, C_k$ of $\mathbf{G}$ such that $i < j$ if $C_i \subseteq \overline{C_j}$.

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$V_{d_1}^1$

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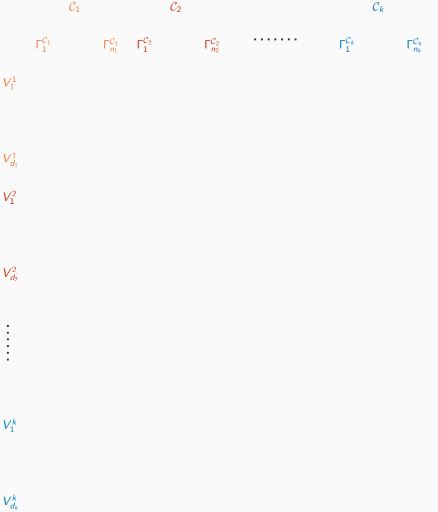

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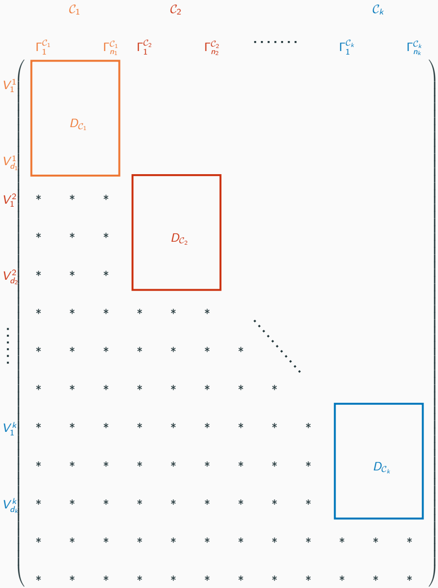

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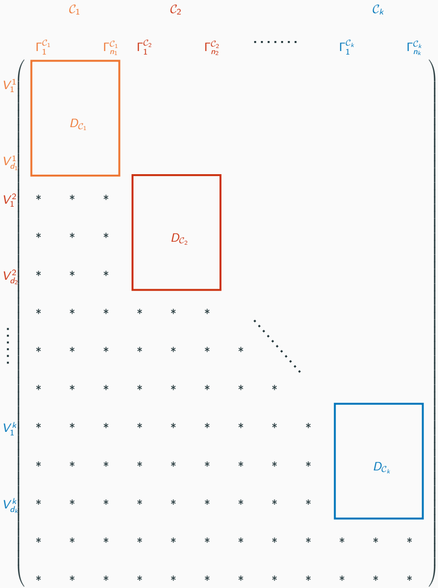

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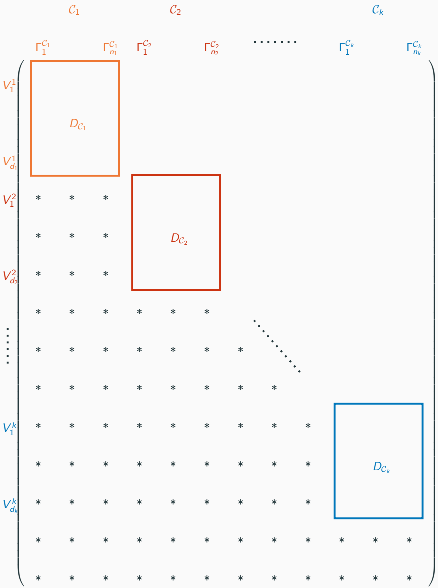

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Solution: decompose the GGGRs into **Kawanaka** modules.

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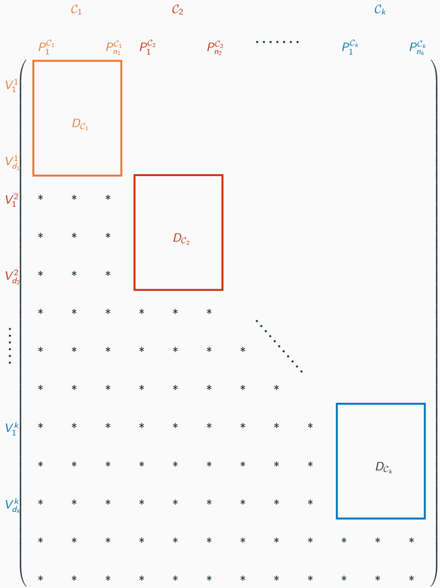

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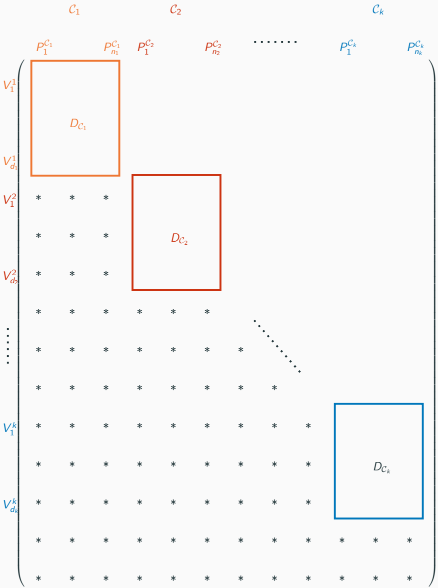

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For each unipotent conjugacy class C, check that the matrix D_C is unitriangular.

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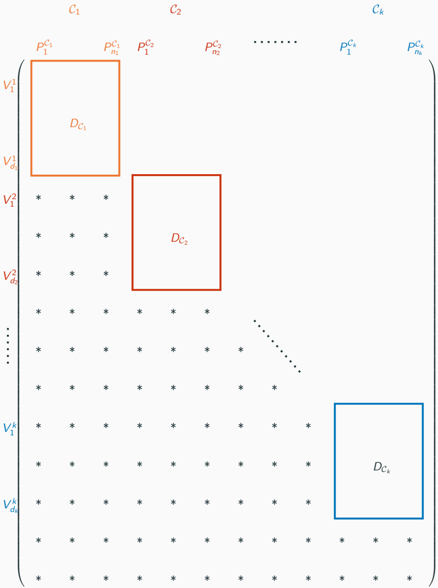

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For each unipotent conjugacy class C, check that the matrix D_C is unitriangular.

We need the characters values of the ordinary modules in the basic set.

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Character sheaves are $\mathbf{G}$-equivariant irreducible perverse sheaves on $\mathbf{G}$.

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Let $\mathcal{A} \in \hat{\mathbf{G}}$. We want the value of $\mathcal{A}_{(su)_\mathbf{G}}$ for $s \in \mathbf{G}$ semisimple, $u \in C^\circ_\mathbf{G}(s)$ unipotent, and $(u)_\mathbf{G}$ is the unipotent support of $\mathcal{A}$.

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This is due to the **generalised Springer correspondence**.

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$\alpha(\mathcal{A}, X)$ is approximately $\langle Y, \mathrm{Res}^{W^{\mathbf{G}}_{\mathbf{L}}}_{W_{\mathfrak{m}}} \mathrm{Ind}^{W^{\mathbf{G}}_{\mathbf{L}}}_{W^{\mathbf{H}}_{\mathbf{L}_0}}(X) \rangle$ with a twist.

Unitriangularity of decomposition matrices

Recall that $\ell \neq p$.

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Assume that $\mathbf{G}$ *is simple exceptional of adjoint type and that p is good for* $\mathbf{G}$*. Then, the* ℓ*-decomposition matrix of the unipotent* ℓ*-blocks of* $\mathbf{G}^F$ *is lower-unitriangular (w.r.t. a suitable ordering).*

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Assume that $\mathbf{G}$ *is simple of type* G_2 *and* F_4 *and that* p *is good for* $\mathbf{G}$*. Then, the* ℓ*-decomposition matrix of the isolated* ℓ*-blocks of* $\mathbf{G}^F$ *is lower-unitriangular (w.r.t. a suitable ordering).*

Thank you!

Comments on the results/proofs

- The condition p good is crucial for the GGGRs.
- Some cases are a consequence of a result of Geck–Hézard (2008).
- In a lot of cases, there is no need to compute the values of all character sheaves at all mixed conjugacy classes, but only at the unipotent support.
- For the isolated blocks, Kawanaka characters are sometimes not enough.
  Solution: apply Harish-Chandra theory and the knowledge that the decomposition matrix of the Levi subgroup is lower-unitriangular.

Next questions to consider

- Unitriangularity of decomposition matrices for the isolated blocks of type E_6, E_7 and E_8.
- Unitriangularity of decomposition matrices in the disconnected centre case.
- Properties of basic sets, e.g. stable under group automorphisms.
- Characteristic functions of character sheaves at mixed conjugacy classes.

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